

大数据计算及应用(一)

课程介绍

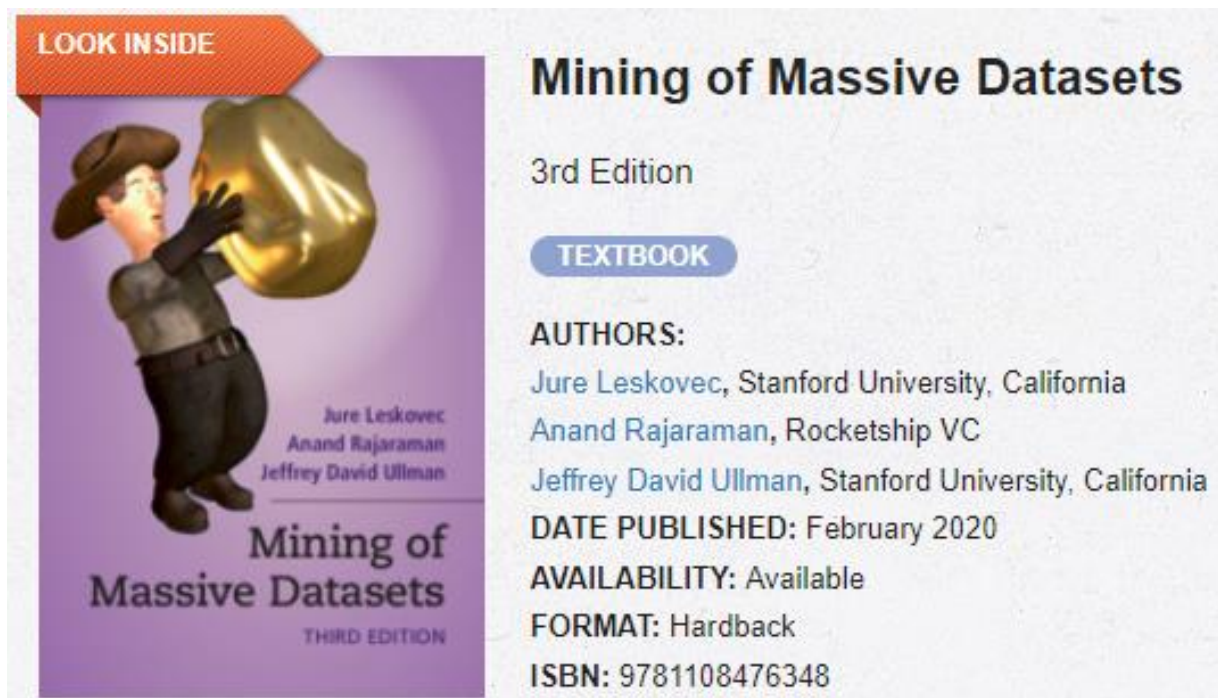
课程基本情况

□ 参考书

英文版: Mining of Massive Datasets

参考资料网址: <http://www.mmds.org/>

中文版: 大数据:互联网大规模数据挖掘与分布式处理



基本情况

□ 具体讲读内容

参考书：大数据挖掘

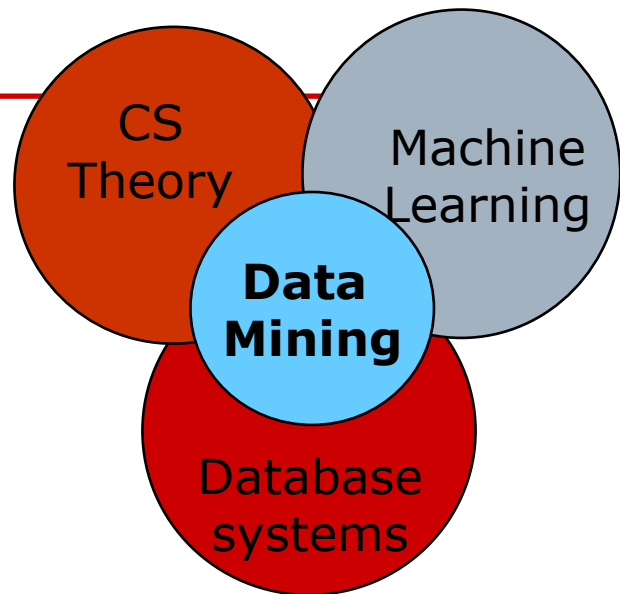
参考资料网址: <http://www.mmnds.org/>

Chapter	Title	Book	Slides	
	Preface and Table of Contents	PDF		
Chapter 1	Data Mining	PDF	PDF	PPT
Chapter 2	Map-Reduce and the New Software Stack	PDF	PDF	PPT
Chapter 3	Finding Similar Items	PDF	PDF	PPT
Chapter 4	Mining Data Streams	PDF	Part 1: PDF Part 2: PDF	PPT PPT
Chapter 5	Link Analysis	PDF	Part 1: PDF Part 2: PDF	PPT PPT
Chapter 6	Frequent Itemsets	PDF	PDF	PPT
Chapter 7	Clustering	PDF	PDF	PPT
Chapter 8	Advertising on the Web	PDF	PDF	PPT
Chapter 9	Recommendation Systems	PDF	Part 1: PDF Part 2: PDF	PPT PPT
Chapter 10	Mining Social-Network Graphs	PDF	Part 1: PDF Part 2: PDF	PPT PPT
Chapter 11	Dimensionality Reduction	PDF	PDF	PPT
Chapter 12	Large-Scale Machine Learning	PDF	Part 1: PDF Part 2: PDF	PPT PPT
	Index	PDF		
	Errata	HTML		

数据挖掘介绍

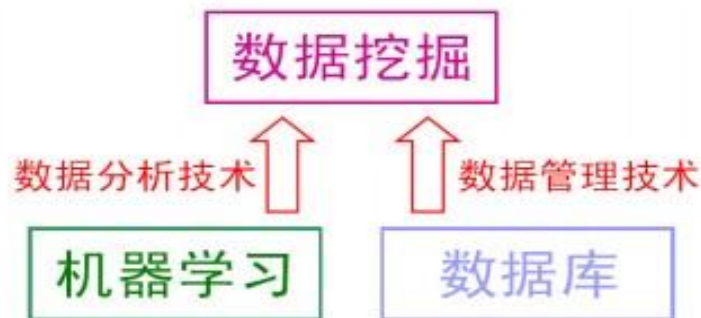
□ 数据挖掘与下列研究领域有关：

- 数据库：大数据，简单查询
- 机器学习：小数据，复杂模型
- 计算机理论：算法



□ 不同点：

- 与数据库研究相比，数据挖掘偏向于数据分析——即如何处理大型数据的复杂查询
- 与机器学习相比，数据挖掘更侧重于结合具体应用，以及借鉴了数据库的数据管理方法



涉及内容

□ 挖掘各类型数据：

- 高维度数据
- 图数据
- 动态数据(流数据)
- 标注数据

□ 运用多种计算模式

- 单机基于内存模式
- 流数据处理算法
- 分布式

涉及内容

□ 学会解决实际问题

- 重复文章查找
- 超市商品关联分析 (Market Basket Analysis)
- 推荐系统 (Recommender systems)
- Spam 检测

□ 学会各工具

- 线性代数(SVD, Rec. Sys., Communities)
- 优化 (stochastic gradient descent)
- 动态编程
- 散列法 (Hashing: LSH, Bloom filters)

总体涉及技术/课题

高维数据

Locality
sensitive
hashing

Clustering

Dimensionality
reduction

图数据

PageRank,
SimRank

Community
Detection

Spam
Detection

流数据

Filtering
data
streams

Web
advertising

Queries on
streams

机器学习

SVM

Decision
Trees

Perceptron,
kNN

应用

Recommen
der systems

Association
Rules

Duplicate
document
detection

课程成绩计算

□ 平时成绩：20%

■ 课上(后)问答题

□ 2个大作业：80%

□ 课件获取：学院网站/微信群

□ 联系邮箱：

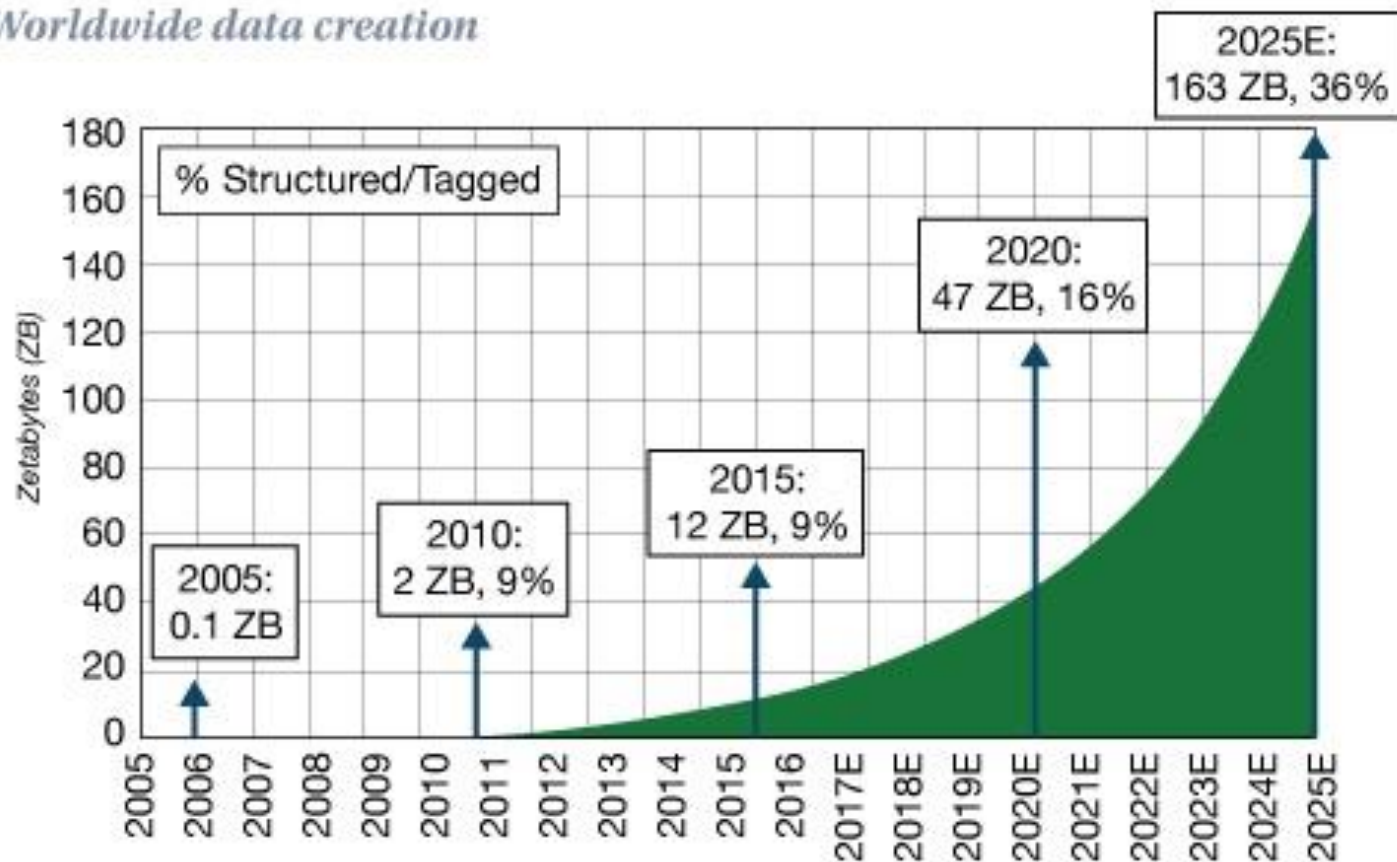
教师：杨征路，邮箱：yangzl@nankai.edu.cn

助教：bigdatacomputing@163.com

时代背景：信息爆炸

Data Explosion

Worldwide data creation



Source: Kleiner Perkins

大数据的特征

□ 数据量大 (Volume)

- 环境，生物数据等

□ 高速 (Velocity)

- 传感器网络，金融股票数据等

□ 多类型 (Variety)

- 文本，声音，视频，图像数据等

"Big Data are **high-volume**, **high-velocity**, and/or **high-variety** information assets that require new forms of processing to enable enhanced decision making, insight discovery and process optimization"

Gartner 2012

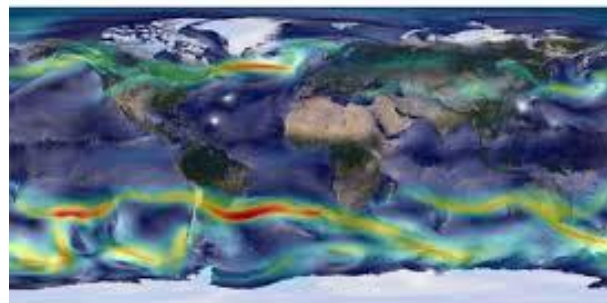
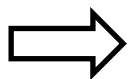
□ 价值 (Value)

多大的数据算大数据？

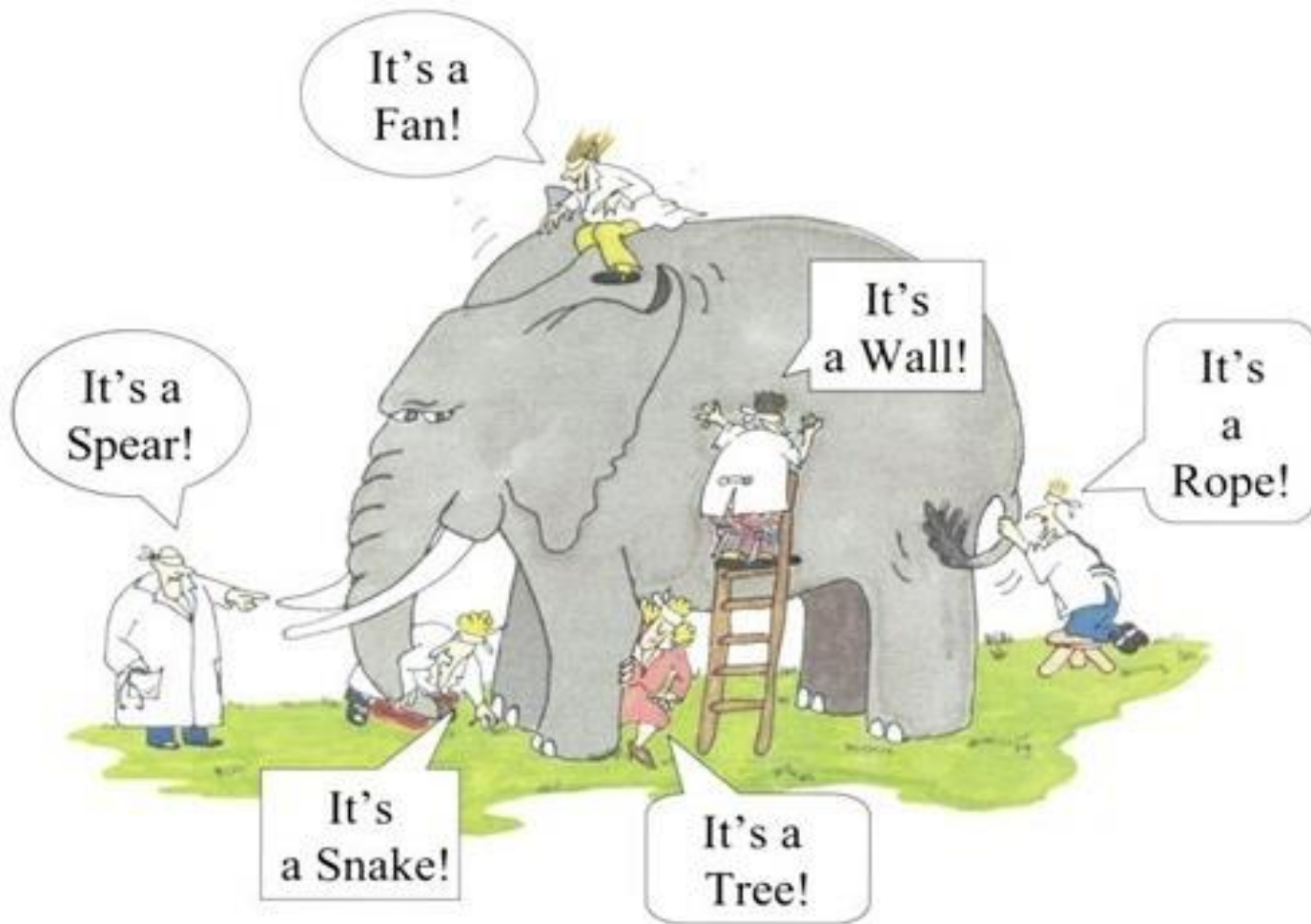
□ 1GB(10^9)? 1PB (10^{15})? 1ZB (10^{21})?

规模巨大到无法通过目前主流软件工具，在合理时间内达到撷取、管理、处理、并整理成为帮助企业经营决策更积极目的的资料(维基百科)

□ 我们需要多大的数据？



我们需要多大的数据？



□ 完整性 (Integrity)

大数据的特征

□ 多类型 (Variety)

- 文本、声音、视频、图像数据等

□ 数据量大 (Volume)

- 环境、生物、网络数据等

□ 高速 (Velocity)

- 传感器网络、金融股票数据等

□ 价值 (Value)

视频片断搜索

- 问题点：对应长视频，用户往往只对其中一段感兴趣



Log in to chat

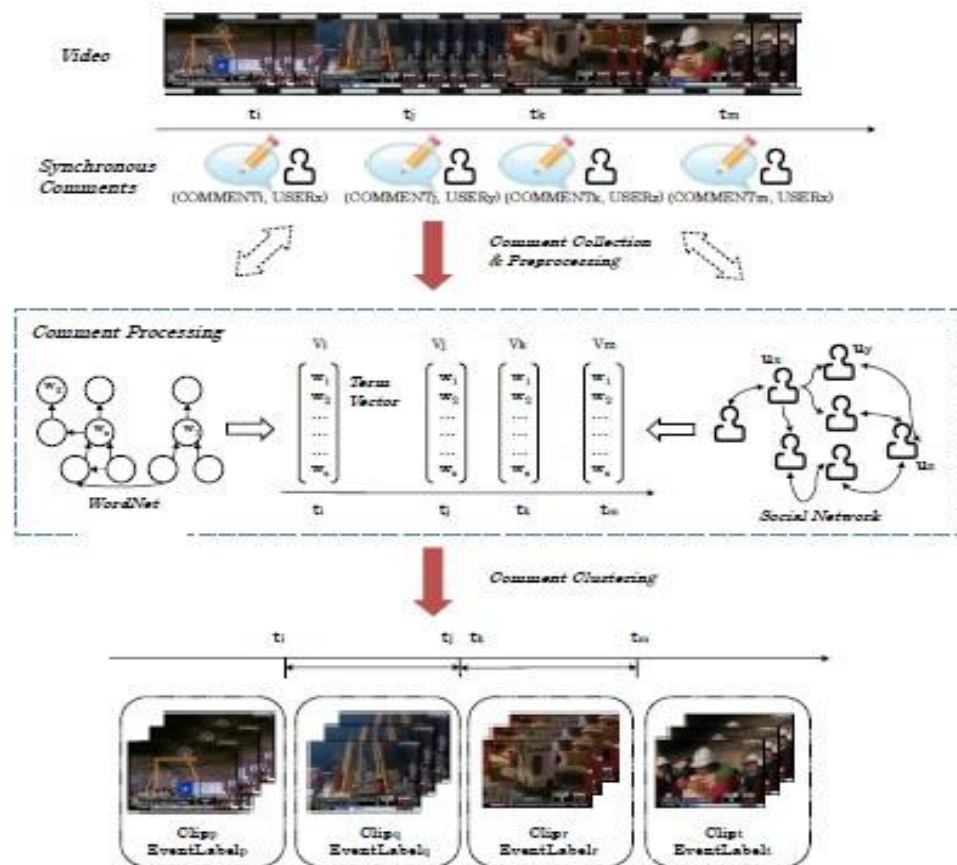
Comment

Twitter user

Timestamp

- 利用文本信息和大众智慧 (Wisdom of Crowd)
- 核心理念：(1) 对评论作cluster (基于时间, 文本还有用户相似度)
- (2) 对应用户的查询, 找出相应的视频cluster

视频片段搜索



- 系统组成：
 - (1) 时间相似度
 - (2) 文本相似度
 - (3) 用户相似度
- 通过交叉验证设置权重

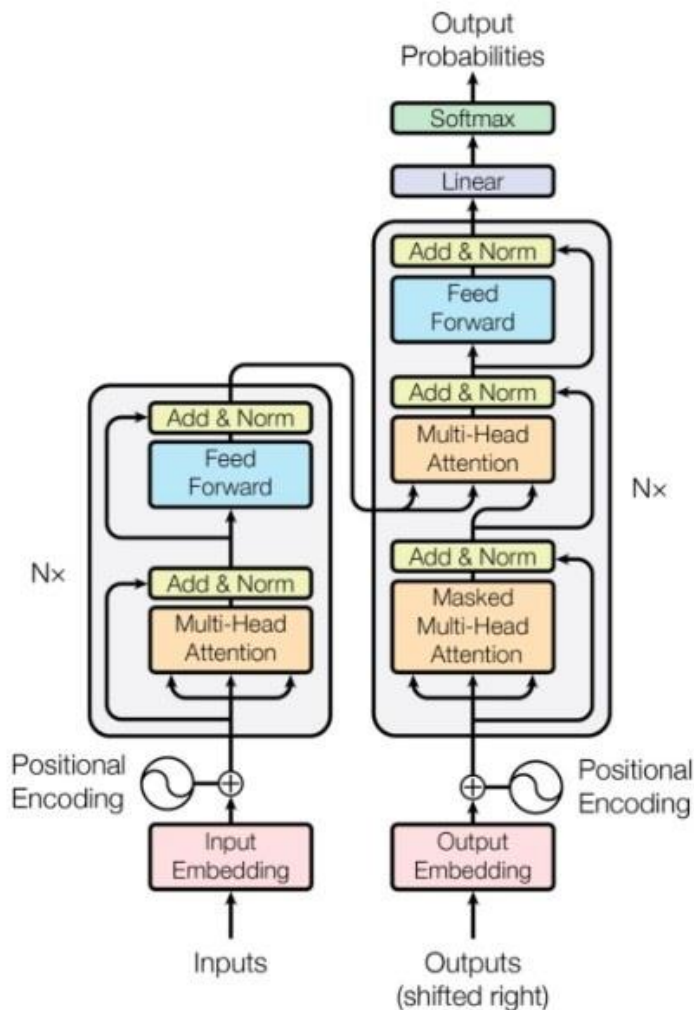
Query ID	BA			SE			CR		
	Prec.	Rec.	F-1	Prec.	Rec.	F-1	Prec.	Rec.	F-1
1	0.476	0.567	0.518	0.307	0.62	0.41	0.54	0.605	0.571
2	0.6	0.366	0.454	0.554	0.463	0.504	0.726	0.488	0.583
3	0.916	0.26	0.405	0.933	0.333	0.491	1	0.396	0.568
4	0.853	0.924	0.887	0.85	0.964	0.903	0.91	0.929	0.919

- 1) BA: 考虑统计信息的简单的向量空间模型
时间度+tf-idf
- 2) SE: 利用基于Wordnet的文本相似度
时间度+语义相似度(Wordnet, tf-idf)
- 3) CR: 利用Wordnet和人际关系网络(Twitter)
时间度+语义相似度+用户相似度

多类型数据应用

□ 翻译系统

猫 $\Leftrightarrow$ Cat



The Transformer - model architecture.

多类型数据应用

□ 翻译系统

猫 $\Leftrightarrow$ Cat



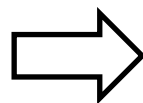
大数据的特征:多类型 (Variety)

□ 难点

- 获取全面的多类型数据
- 最近趋势: 可穿戴设备, 如Google眼镜等



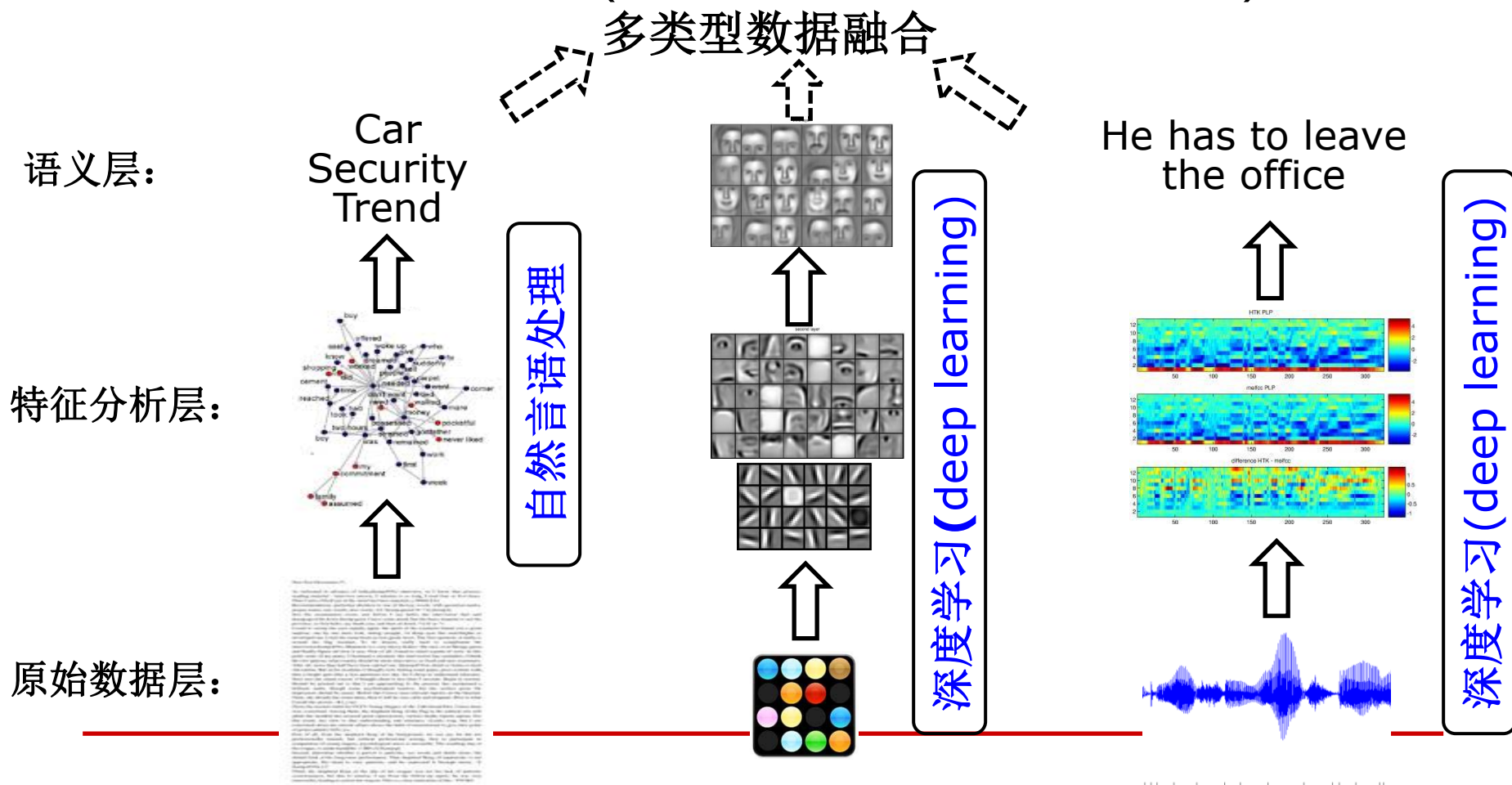
- 巧妙的方式: reCAPTCHA



大数据的特征:多类型 (Variety)

□ 难点

- 获取全面的多类型数据
- 融合多类型数据(文本、语音、图像、视频等)



大数据的特征

□ 多类型 (Variety)

- 文本、声音、视频、图像数据等

□ 数据量大 (Volume)

- 环境、生物、网络数据等

□ 高速 (Velocity)

- 传感器网络、金融股票数据等

□ 价值 (Value)

大数据的特征: 数据量大(Volume)

□ 搜索应用: 例如在Google中输入“Big data”

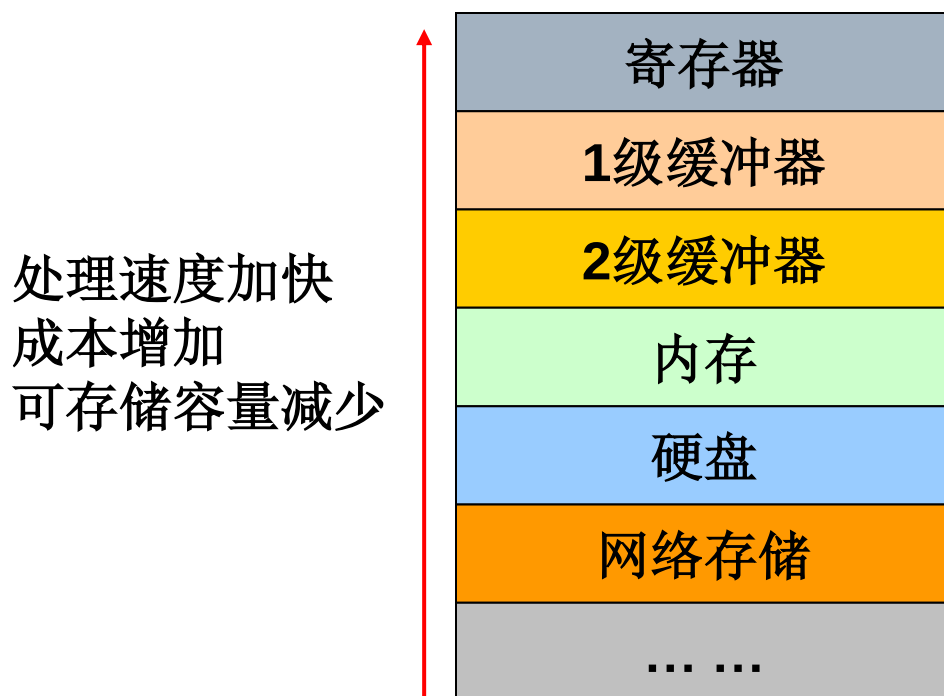


大数据的特征: 数据量大(Volume)

□ 难点

■ 效率问题

➤ 需要结合计算机硬件特点



(1) 尽量把所有数据放在内存里 (128G-1T)

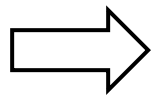
效率优化手段: 减少不必要的数据搜索(或比较及计算)

基于内存计算

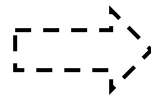
□ 当前服务器的内存

■ 128G 到 1T

1993年
内存: 128K



2023年
内存: 128G



2053年
内存: 128P (2^{50})

大数据的特征: 数据量大(Volume)

□ 难点

■ 效率问题

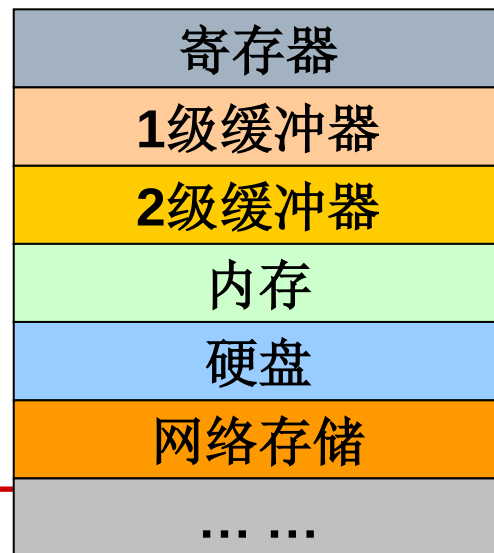
➤ 需要结合计算机硬件特点

(1) 把所有数据放在内存里

(2) 如果数据放不进内存, 则放进硬盘存储

效率优化手段: 减少不必要的硬盘读写数据操作(I/O)

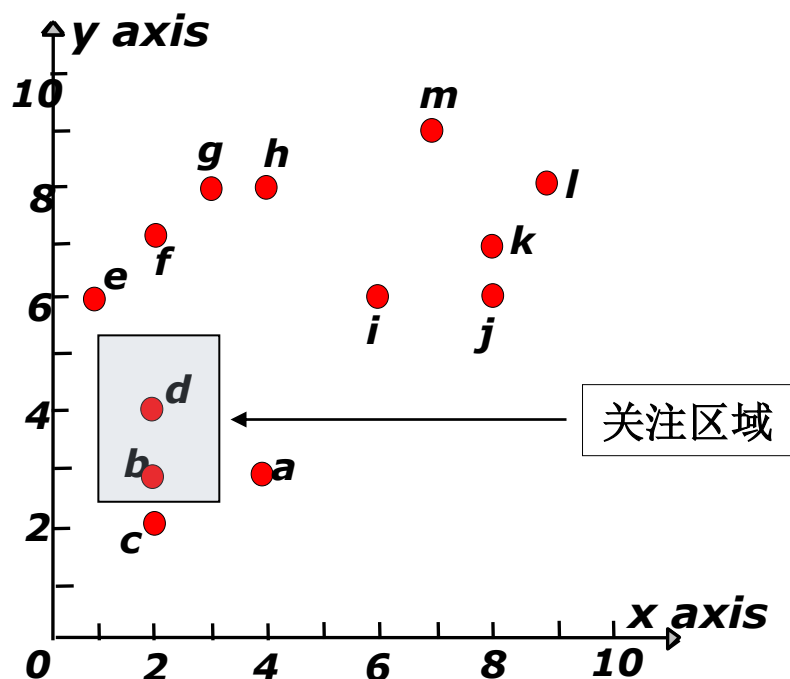
处理速度加快
成本增加
可存储容量减少



基于硬盘的高效计算

□ 有效的数据结构

■ R-tree, kd-tree等



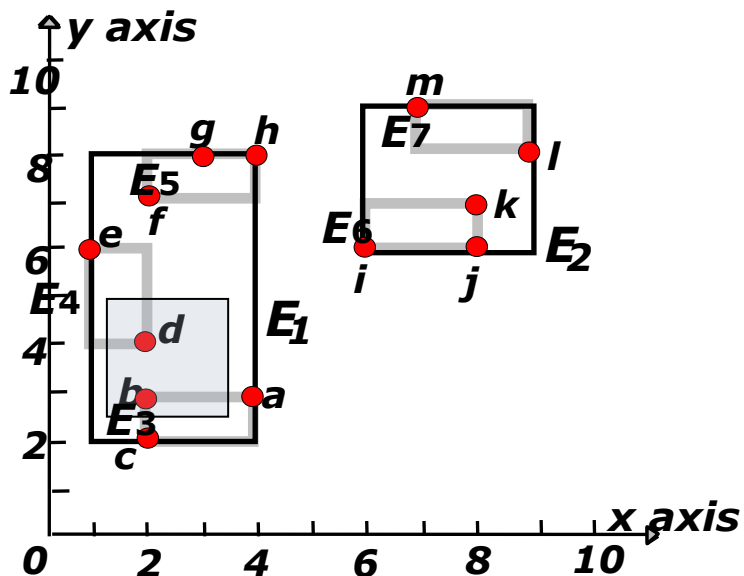
查询：“找到所关注区域内所有的旅馆”

简单方法：逐一扫描硬盘上所有的点信息，计算并得到结果

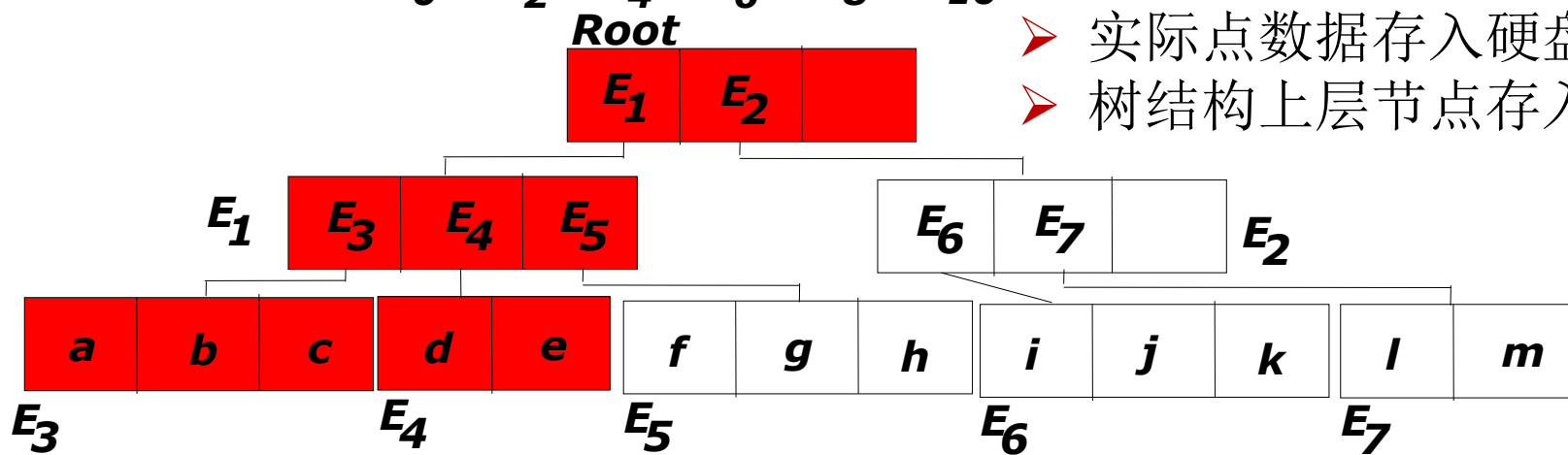
基于硬盘的高效计算

□ 有效的数据结构

■ R-tree



- 实际点数据存入硬盘
- 树结构上层节点存入内存



大数据的特征: 数据量大(Volume)

□ 难点

■ 效率问题

➤ 需要结合计算机硬件特点

(1) 把所有数据放在内存里

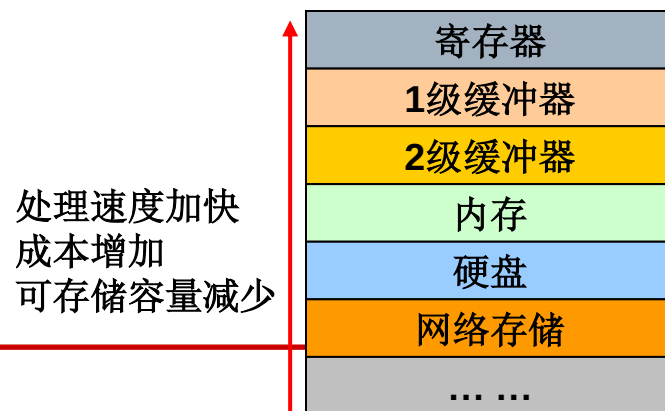
(2) 如果数据放不进内存, 则放进硬盘存储

(3) 尽量增加新的硬件设备, 如

GPU (图形处理单元): 可做并行处理器

SSD (固态硬盘): 读数据速度 >> 写数据速度

效率优化手段: 增加SSD读操作, 减少写操作



大数据的特征: 数据量大(Volume)

□ 难点

■ 效率问题

➤ 需要结合计算机硬件特点

(1) 把所有数据放在内存里

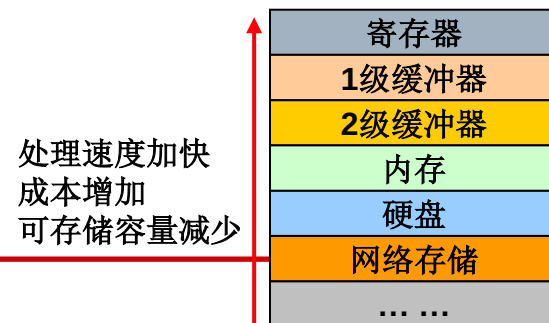
(2) 如果数据放不进内存, 则放进硬盘存储

(3) 增加新的硬件设备

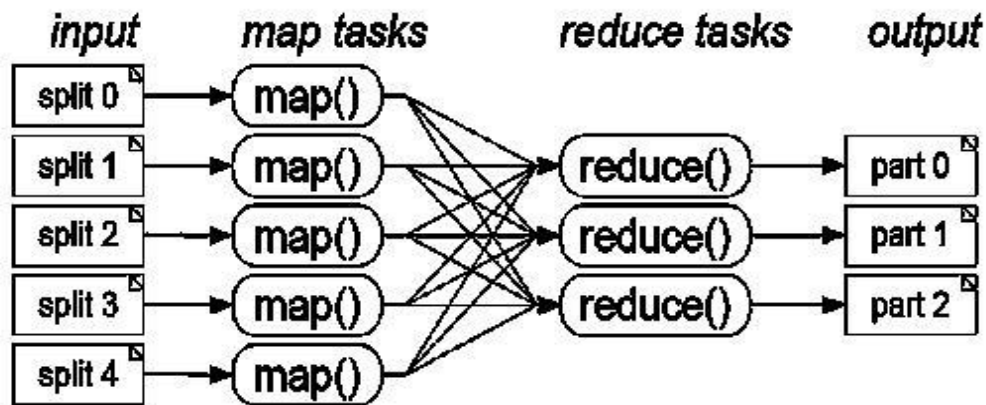
(4) 计算集群 (Computer Cluster)

如 MapReduce (Hadoop等), Spark (由UC Berkeley开发)

效率优化手段: 因网络带宽为分布计算瓶颈, 所以应
少移动数据, 而应该尽量在本地计算



MapReduce计算流程



□ 优点

- 配置及运行容易(只需设计Map和Reduce两个function)

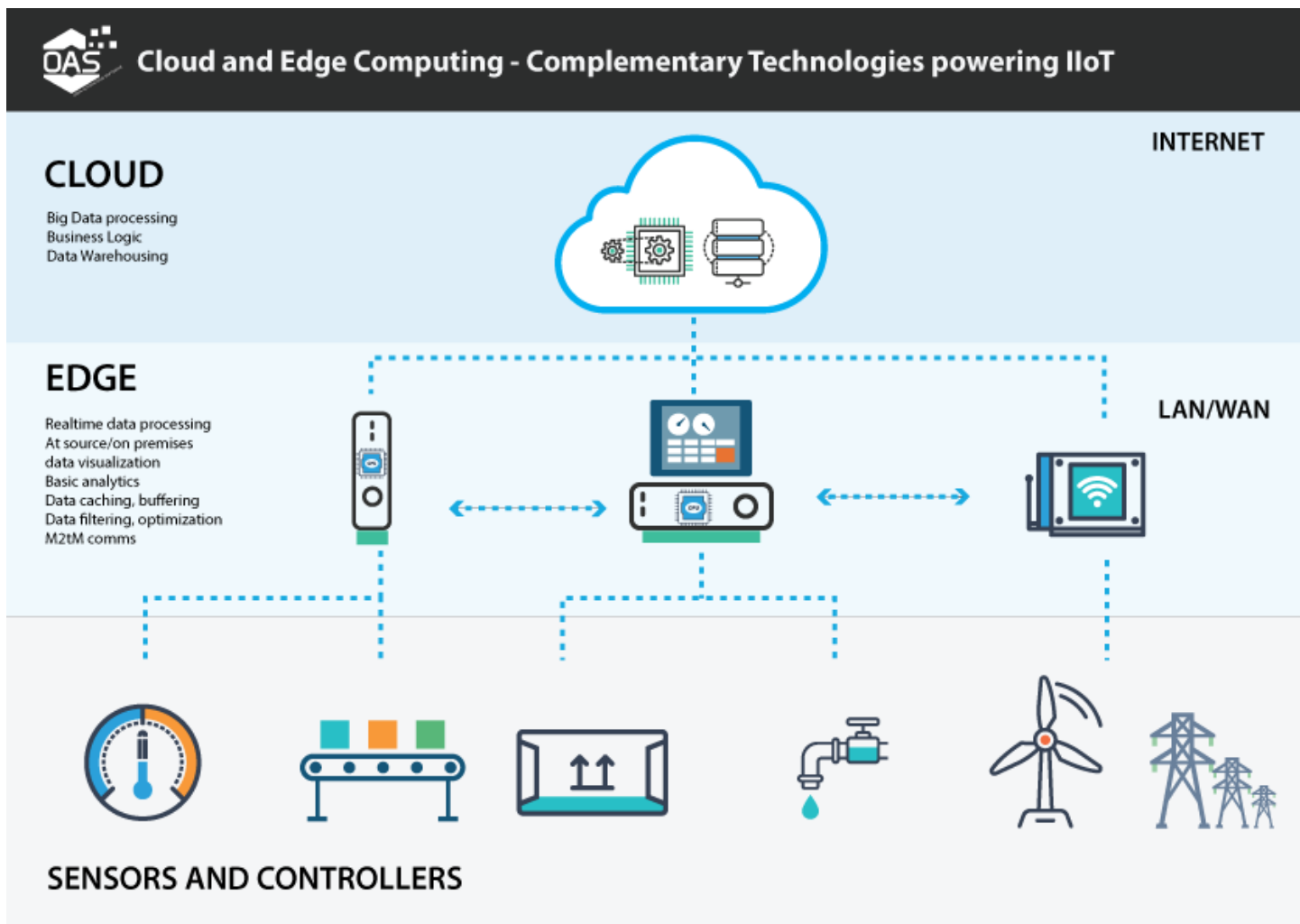
□ 缺点

- 无法控制底层计算机节点
- 用硬盘存储中间数据，效率低
- 无法执行迭代多的任务

□ Spark系统 (UC Berkeley)

- 基于内存计算
- 可调控底层计算机节点

边缘计算 (Edge Computing)



- ❑ 边缘运算将服务加以分解后分散到边缘节点去处理。边缘节点更接近于用户终端装置

大数据的特征

□ 多类型 (Variety)

- 文本、声音、视频、图像数据等

□ 数据量大 (Volume)

- 环境、生物、网络数据等

□ 高速 (Velocity)

- 传感器网络、金融股票数据等

□ 价值 (Value)

大数据的特征: 高速(Velocity)

□ 处理方法

- 按滑动窗口切割数据流 (时间窗大小取决于硬件设备、处理事务类型等)
- 按静态数据处理各时间窗内数据

□ 可优化手段

- 过滤、采样、增量索引等

□ 现有开源系统

- Storm (Twitter公司)
- S4 (Yahoo公司)

大数据的特征

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- 传感器网络、金融股票数据等

□ 价值 (Value)

推荐系统

amazon.com

Google

Top Holiday Deals
In Sports & Outdoors
See today's deals

Browse

Featured Holiday Links
Holiday Sales & Deals in Sports & Outdoors Gift Guide

2012 Editors' Picks
Featured Fitness Links

Clothing
Footwear
Fitness & Yoga DVDs
Items eligible for FREE Super Saver Shipping and Amazon Prime

Accessories

Exercise Balls
Bands
Jump Ropes
Mats
Stopwatches
All Accessories

Cardio Training

Bike Trainers
Elliptical Trainers
Exercise Bikes
Rowers
Step Machines
Treadmills
All Cardio Training

Exercise & Fitness

Find treadmills elliptical trainers home gyms athletic apparel heart rate monitors yoga gear and more

Great Fitness Gifts

- Shop Fitness DVDs
- Shop Weights & Benches
- Shop Fitness Technology
- Shop Yoga Essentials
- See more



Best-Selling
Ellipticals
Shop now

Holiday
Editors' Picks
Shop now

New Trend:
Office Fitness
Shop now

Seasonal Fitness Markdowns



Top Stories

Miami Heat
International Lease
Finance Corporation
HSBC
Mali
X-37B
Google Maps
Tom Brady
Golden State Warriors
Zack Greinke
Prostate cancer

Tokyo, Japan

World

U.S.

Business

Technology

Entertainment

Sports

Science

Health

Top Stories



See
realtime
coverage

Commentary: Right to work will bring jobs home

The Detroit News - 54 minutes ago
Gov. Rick Snyder and legislative leaders embarked last week on the bumpy journey to make Michigan a right to work state. If they succeed, they will vault Michigan to the top tier of states competing for new business and more jobs.

Opinion: Worker Liberation in Michigan Wall Street Journal
In Depth: Michigan's Right to Work Law Washington Times

Related Rick Snyder » Trade union » United Auto Workers »



The fiscal cliff's whale of a tale

Albany Times Union - 58 minutes ago
A "grand bargain" to avert the fiscal cliff and defuse our growing debt bomb must include tax increases and spending cuts.

Navy identifies SEAL killed in hostage rescue in Afghanistan

Reuters - 40 minutes ago
Tue Dec 11, 2012 1:23am EST. (Reuters) - A U.S. Navy SEAL was killed in Sunday's rescue mission in Afghanistan that freed an American doctor kidnapped by the Taliban, the Defense Department said on Monday.

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Recent

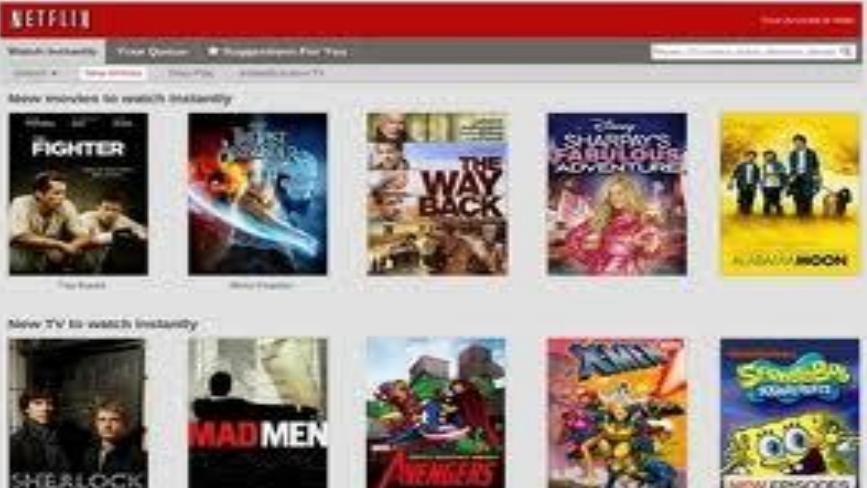
Nikkei eases, investors cautious after recent sharp rally
Reuters - 13 minutes ago

Mexican president confident of key reforms in 2013
Reuters - 36 minutes ago

AP Exclusive: ACLU seeks OAS probe of Padilla case
Houston Chronicle - 57 minutes ago

Tokyo, Japan » - Change location

NETFLIX



lost.fm

People listening right now:

- thesoozbutton is listening to Geographer (indie pop, electronic, indie)
Scrobbling from Spotify
- scot77 is listening to Shawn Colvin (female vocalists, singer-songwriter, folk)
Scrobbling from iTunes in United Kingdom
- JayLady is listening to Fefe Dobson (pop, female vocalists, rock)
Scrobbling from iTunes
- Stritoh is listening to Metallica (thrash metal, metal, heavy metal)
Scrobbling from The Last.fm Scrobbler in Chile
- kunprof is listening to Coldplay (rock, alternative, britpop)
Scrobbling from The Last.fm Scrobbler in Russian Federation

Artists & Labels

Join your fans, upload your music, and earn royalties. Learn more »

Developers & Designers

Looking for resources? Grab a logo pack or head over to our API »

Last.fm recommendations give you:



Music

Recommendations based on what you love with a little twist of something different.

Radio

Listen to endless personalised radio stations based on an artist, tag — or even a friend's taste!

Concerts

Never miss another gig. Based on your taste, Last.fm recommends you events and festivals.

People

Get connected with 'musical neighbours' - people who love the same music as you.

用户和物品数据

用户数据:

(1) 结构化数据

国籍、性别、年龄、兴趣等

(2) 文本型数据

个人描述等

(3) 用户间数据

人际网络等



物品数据:

(1) 结构化数据

价格、重量、颜色、品牌等

(2) 文本性数据

产品描述等

(3) 物品间数据

物品的分类结构树/图

人与物品的交互数据:

(购买, 评价, 网页浏览, 书签标记等)

推荐系统

- 基于协同过滤手法(利用用户和物体之间的关系)
 - 近邻算法 (Nearest Neighbor)
 - 矩阵分解法 (Matrix Factorization)
 - 基于内容的手法(利用用户或物体的特征)
 - 文本相似法 (Text Similarity)
 - 向量空间模型 (Vector Space Model)
 - 基于语义资源模型 (Wordnet, Wiki等)
 - 聚类 (cluster)
 - K-means
 - 层次聚类 (Hierarchical Clustering)
 - 分类 (classification)
 - 决策树 (Decision Tree)
 - 人工神经网络 (Artificial Neural Network)
 - 支持向量机 (Support Vector Machine)
 - kNN
 - Naive Bayes
-

推荐系统

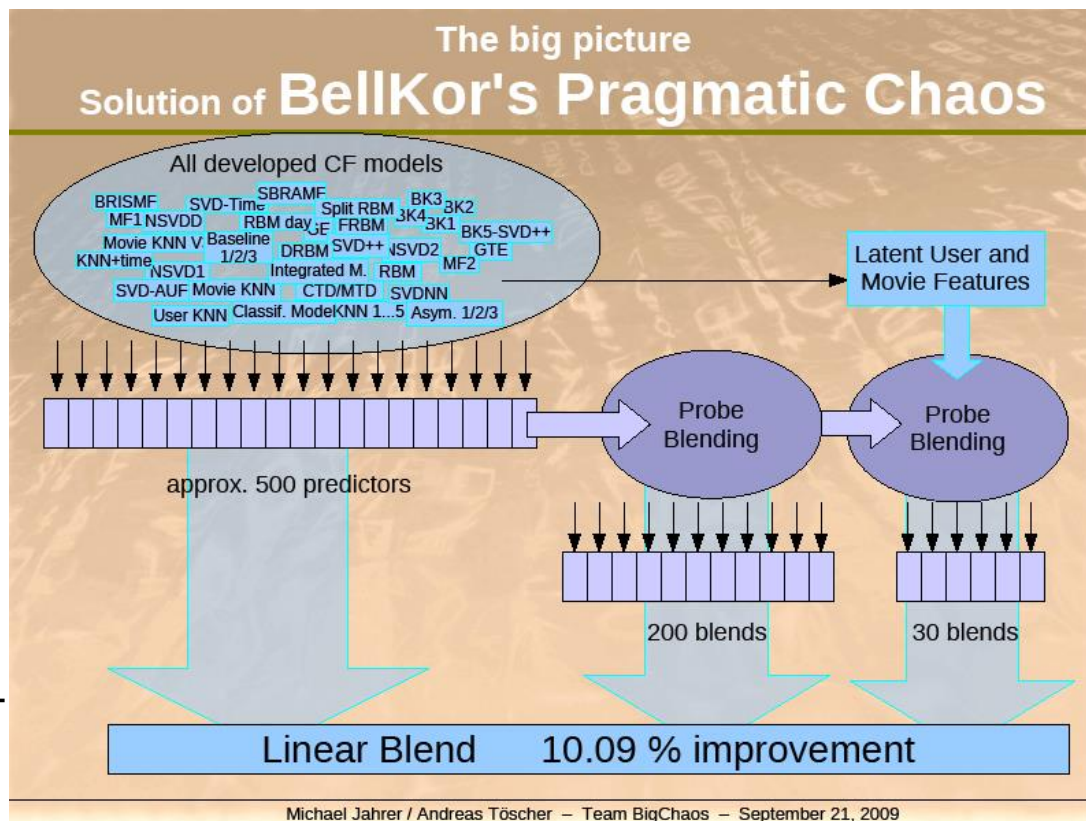
□ 难点

- 如何选择模型
- 如何调整参数

□ 成功案例

Netflix 电影推荐竞赛

- 数据统计量
 - 17,770 部电影
 - 480K 人
 - 100M 评价
 - 6 years of data: 2000-2005
 - 3M测试集
- 2700+ 比赛队伍
- 奖励\$1M 为提高10% 精度



Michael Jahrer / Andreas Töschler – Team BigChaos – September 21, 2009

推荐系统

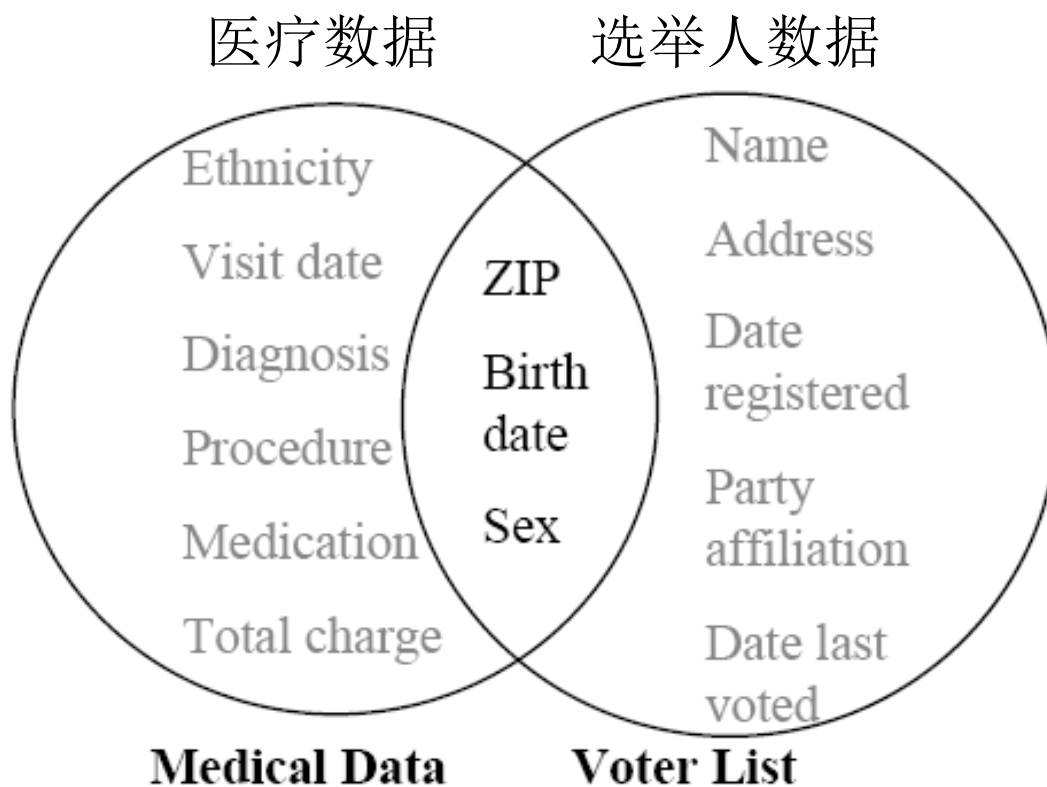
□ 新闻推荐系统

- 今日头条
- 创始人: 张一鸣
南开 软件学院 (2005)
- 创立时间: 2012年8月
- 用户人数: 3.5亿



大数据的隐私保护

Sweeney 在**2001**年发现可以通过网上发布的美国选举人的个人数据，和州政府雇员的医疗数据，来联系推测出政府官员的医疗信息



公开数据库中的个人隐私

名字	生日	性别	邮编	疾病
Person 0001	1962/1/18	Male	08901	癌症
Person 0002	1990/7/22	Male	08904	-----
Person 0003	1969/5/17	Female	08854	哮喘
Person 0004	1947/2/21	Female	08904	癌症
Person 0005	1939/8/16	Male	08854	糖尿病

公开数据库中的个人隐私

名字	生日	性别	邮编	疾病
Joe Smith	1962/1/18	Male	08901	癌症
John Doe	1990/7/22	Male	08904	-----
Mary Johnson	1969/5/17	Female	08854	哮喘
Janie McJonno	1947/2/21	Female	08904	癌症
Johnny Walker	1939/8/16	Male	08854	糖尿病

根据统计估算，**87%**的美国人可以由生日，性别和邮编唯一确定。

大数据的隐私保护

- 有效方法:
 - 干扰 (Perturbation)
 - 交换 (Swapping)
 - 泛化 (Generalization)
 - k -匿名
 - l -diversity
 - t -closeness

K-匿名

Name	Age	Gender	Zip Code	Disease
Person 0001	<32	Male	08901	Cancer
Person 0002	<24	Male	08904	-----
Person 0003	>59	Female	08854	Asthma
Person 0004	<45	Male	08904	Diabetes
Person 0005	>76	Female	08854	Cancer
Person 0006	>61	Female	08854	AIDS

大数据应用

智能医护



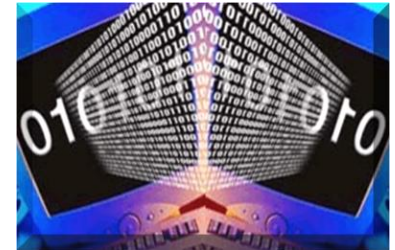
零售



金融



日志分析



Homeland Security



交通控制



通讯



搜索



制造



交易分析

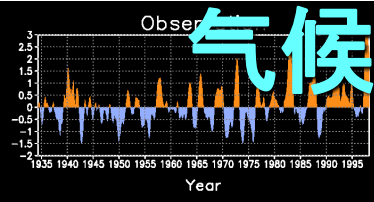
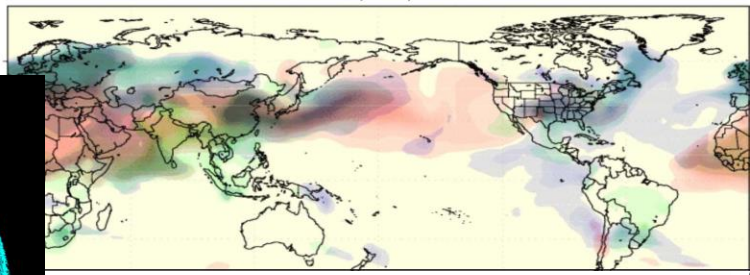
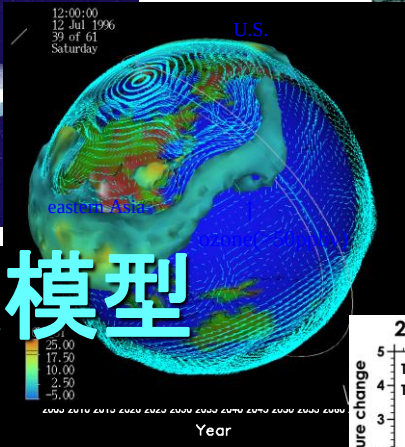
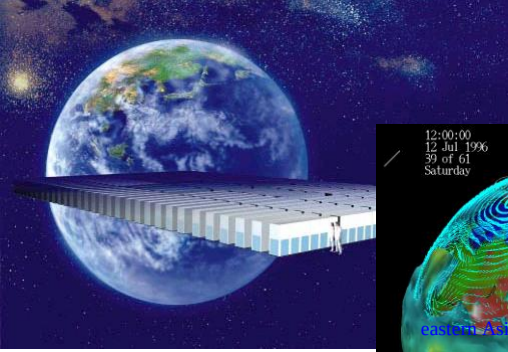


欺诈风险评估

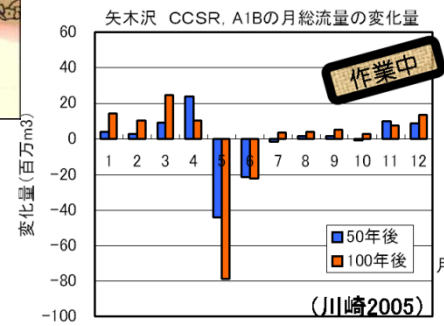
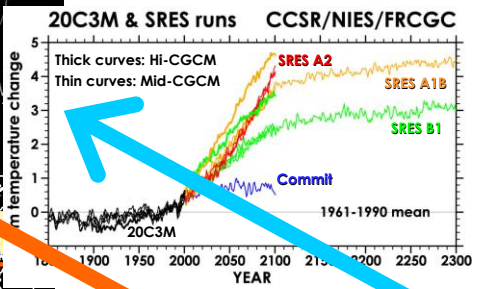
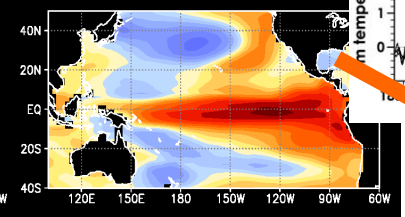
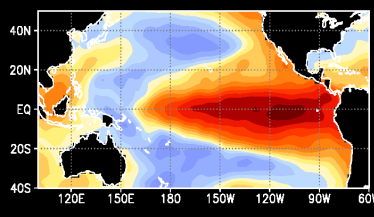


其他数据驱动应用

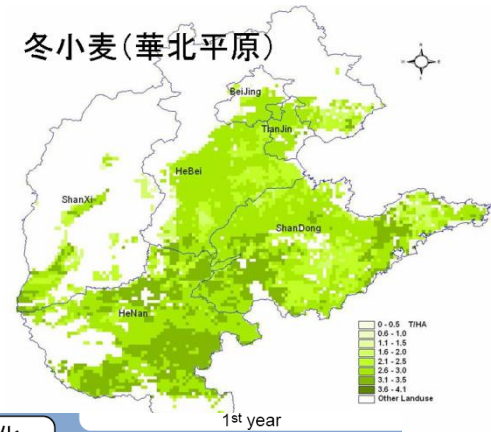




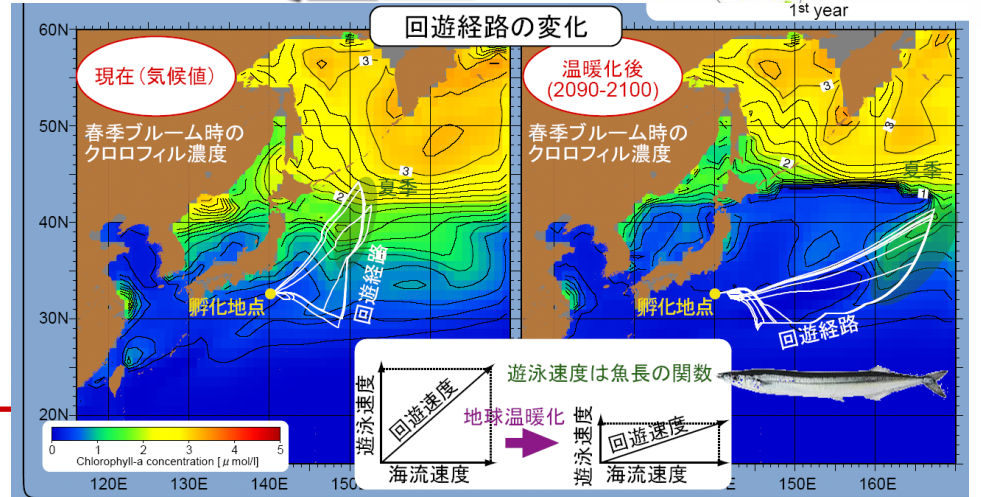
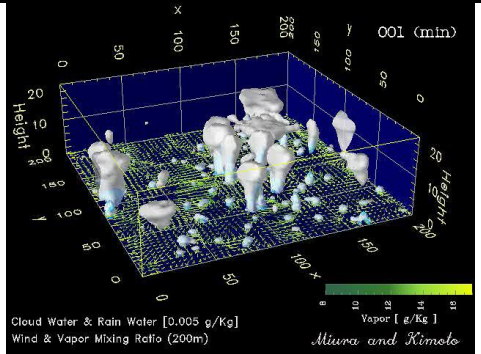
気候模型



冬小麦(華北平原)



农业&渔业



天气数据整合和分析系统（东京大学）

大容量和多样性

数据互用性

存储能力 PB 级

长周期数据管理

数据挖掘

数据整合

天气

模型建立

水循环

生态系统

Ontology

Land utilizing data
schemer

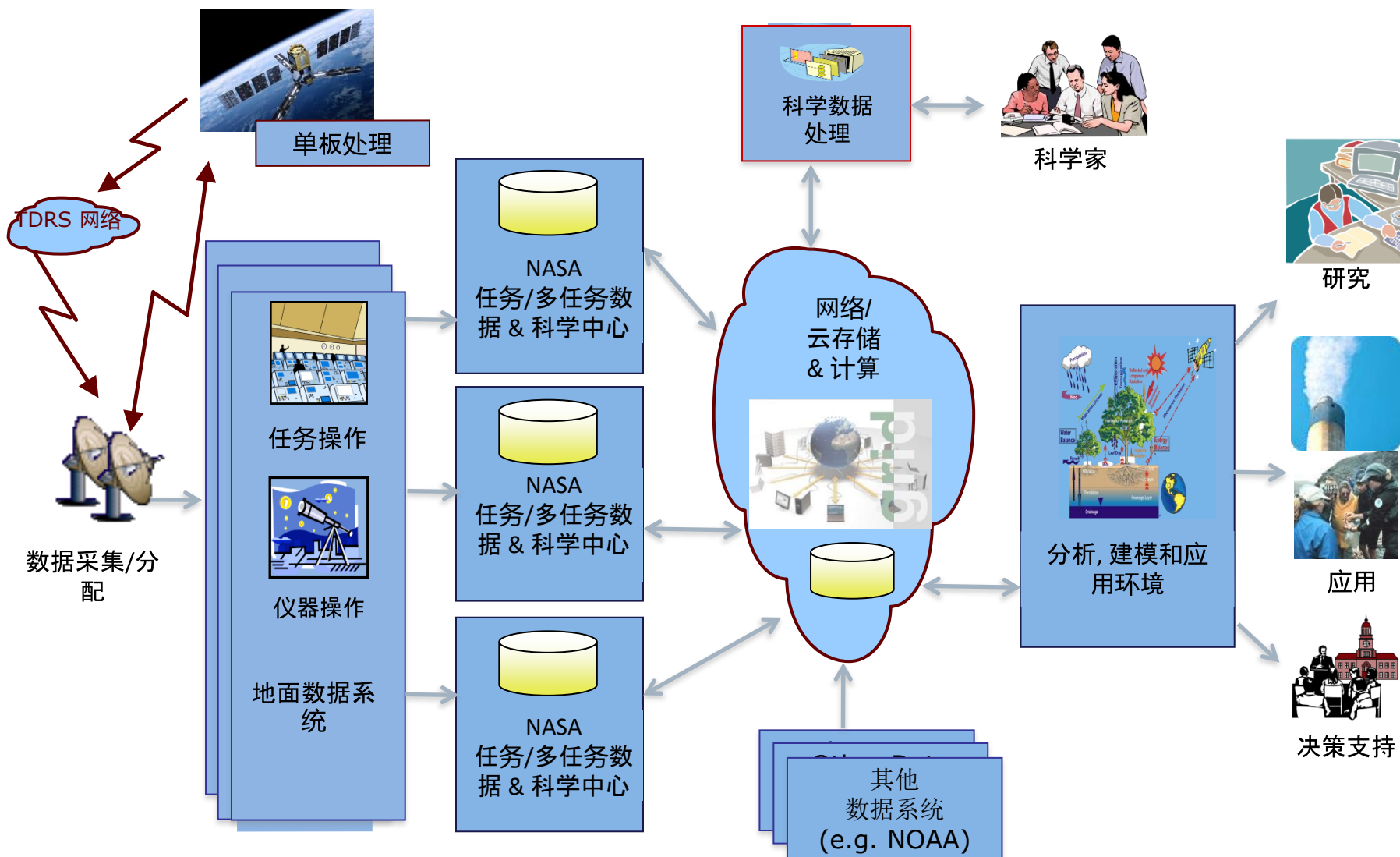
Water cycle data
schemer

Insurance and sanitary data schemer

Metadata Schemer

高级的用户界面

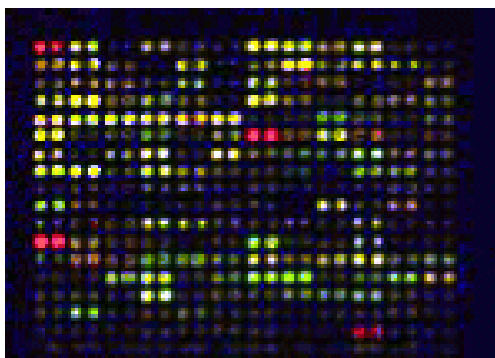
天气数据分析显示及应用（美国NASA）



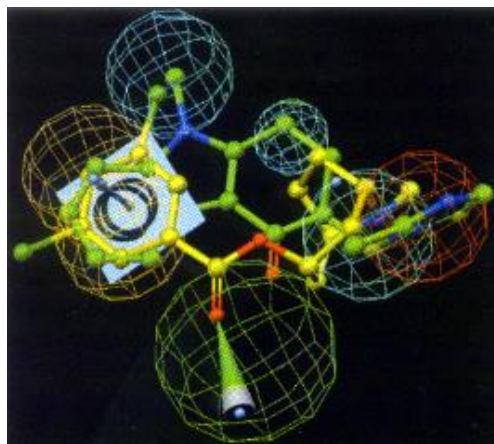
生物数据挖掘与显示

- 2014年财年美国NIH启动利用生物医学大数据计划
 - ✓ 从大数据到知识(big data to knowledge)
- 每年全球产生生物数据约1EB(10^{18})
- 1个人的基因信息(约6GB)

基因序列



结构



功能(疾病)

肥胖症
糖尿病

。 。 。

1984 First public discussion of sequencing the human genome

1985 PCR invented

1986 First automated DNA sequencing instrument developed

1987 First-generation human genetic map developed

1988 Human Genome Organization (HUGO) formed

1989 Cystic fibrosis gene identified by positional cloning

1990 Human Genome Project (HGP) launched in the U.S.

1991 First U.S. Genome Centers established

1992 Second-generation human genetic map developed

1993 New five-year plan for the HGP in the U.S. published

1994 HGP's human genetic mapping goal achieved

1995 HGP's human physical mapping goal achieved

1996 First human gene map established

1997 DOE forms Joint Genome Institute

1998 Incorporation of 30,000 genes into human genome map

1999 Full-scale human sequencing begins

2000 Draft version of human genome sequence completed

2001 Draft version of human genome sequence published

2002 Draft version of mouse genome sequence completed and published

2003 Finished version of human genome sequence completed

1985 Muscular dystrophy gene identified by positional cloning

1986 International Nucleotide Sequence Database Consortium formed

1987 National Research Council (U.S.) issues report on Mapping and Sequencing the Human Genome

1988 Development of yeast artificial chromosome (YAC) cloning

1989 Sequence-tagged sites (STS) mapping concept established

1990 Mendel discovers laws of genetics 1865

1990 Rediscovery of Mendel's work 1900

1990 Gerold formulates the concept of human inborn errors of metabolism 1905

1990 Starveit makes the first linear map of genes 1913

1990 Avery, McClelland, and McCarty demonstrate DNA is the hereditary material 1944

1990 Watson and Crick describe the double helical structure of DNA 1953

1990 Hirshberg, Khosla, and Holley determine the genetic code 1966

1990 Cohen and Boyer develop recombinant DNA technology 1972

1990 Issing of Belmont Report on the use of human subjects in research 1974

1990 Sanger and Maxam & Gilbert develop DNA sequencing methods 1977

1990 GenBank database established 1982

1990 First human disease gene mapped with DNA markers - Huntington disease 1983

1990 First human genome map 1984

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1990 First human genome map 2089

✓ 但90%以上的人类基因尚不了解功能

✓ 如何由数据(Data)变成知识(Knowledge)

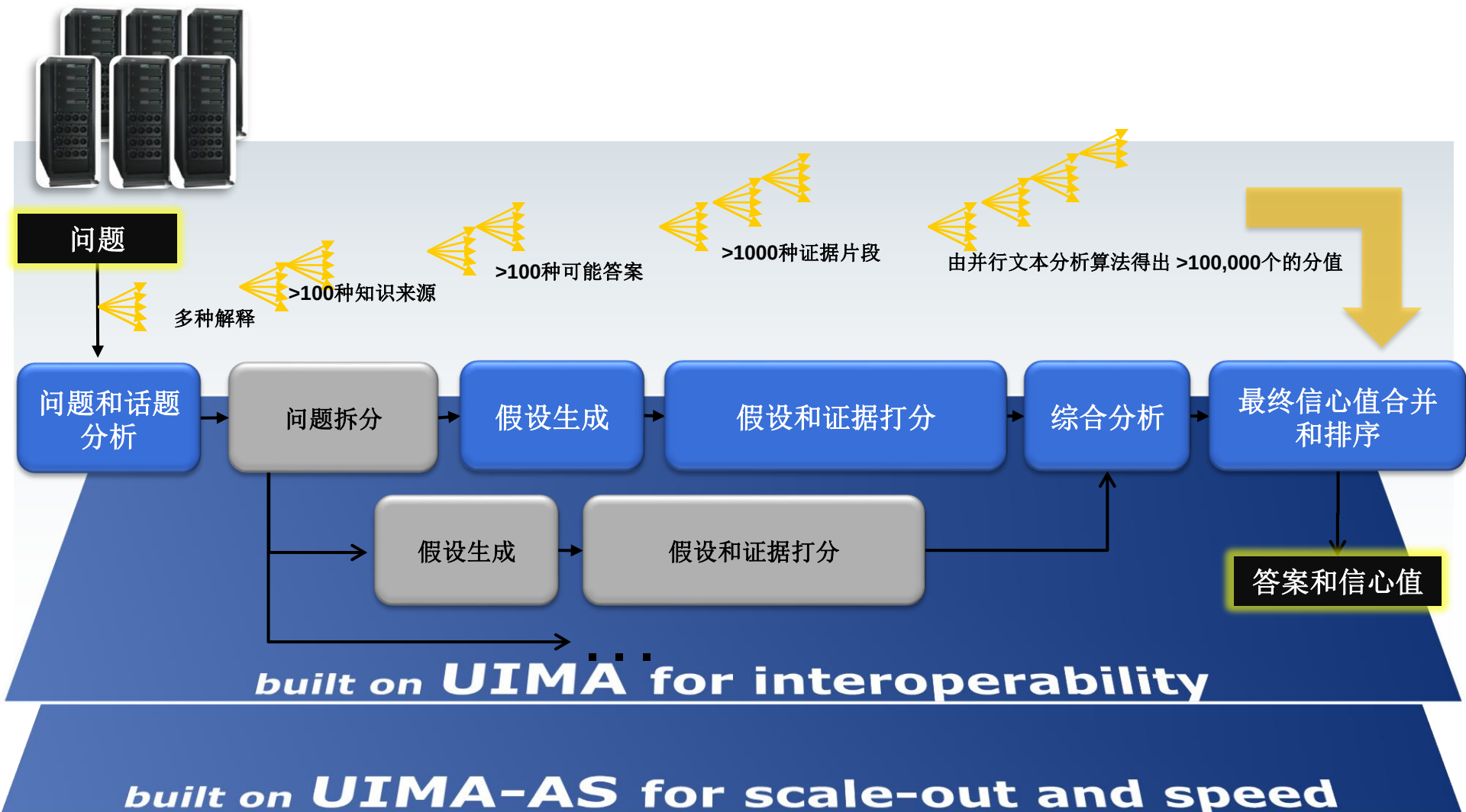
问答系统 – IBM's Watson

➤ Jeopardy (危险边缘)

- ✓ 哥伦比亚广播公司益智问答游戏节目
- ✓ 涉及到历史、文学、艺术、流行文化、科技、体育、地理、文字游戏等等各个领域
- ✓ 2011年2月14号，IBM的沃森系统(Watson)和两名人类冠军选手比赛



一个Jeopardy! 问题在一台2.6G赫兹的机器上要跑2个小时
利用并行系统和非结构化信息管理框架，Watson在2-6秒内给出答案.



流感趋势预测



Tip: Use commas to compare multiple search terms.

Examples

[slashdot.digg](#)

[cat.dog.fish.bird](#)

[denmark.cartoons](#)

[hersheys.com.godiva.com](#)

[kexp.org.kuow.org.kplu.org.kbcs.fm](#)

[ebay.com.craigslist.org](#)

Hot Searches (USA)

- | | |
|---|---------------------------------|
| 1. ellie.goulding | 6. rajon.rondo |
| 2. boxing | 7. fathers.day |
| 3. boxing.news | 8. free.ecards |
| 4. rondo | 9. pay.per.view |
| 5. scott.pilgrim.vs.the.world | 10. snl |

[More Hot Searches »](#)

Explore advanced features with [Google Insights for Search](#)

Explore the audiences of specific websites with [Trends for Websites](#)

[Google Home](#) - [About Google Trends](#)

Google 预测流感趋势

google.org Flu Trends

Language: English (United States)

[Google.org home](#)

Flu Trends

Home

Select country/region

[How does this work?](#)

[FAQ](#)

Flu activity

Intense

High

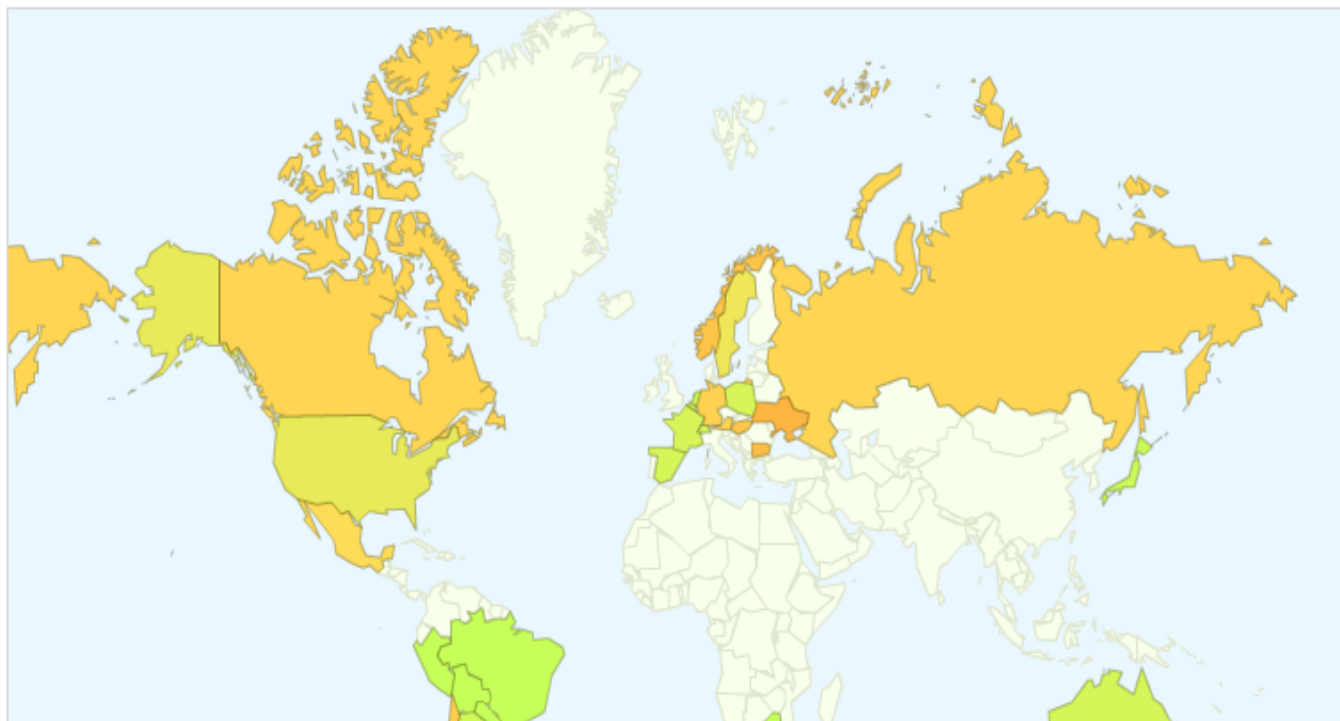
Moderate

Low

Minimal

Explore flu trends around the world

We've found that certain search terms are good indicators of flu activity. Google Flu Trends uses aggregated Google search data to estimate flu activity. [Learn more »](#)



Google 预测流感趋势

- 2008年，Google公司开发了预测世界各地流感发生趋势的软件
 - 主要想法是利用各地Google查询数据来实时评估当前流感发生情况
 - 比美国疾病控制与防御中心提前**10**天预知美国部分州流感发生状况
-

票价趋势预测系统 (Farecast)

[Web](#) [Images](#) [Videos](#) [News](#) [More](#) | [MSN](#) [Hotmail](#) [Sign in](#) [International](#) [Preferences](#)



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RESOURCES

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[Farecast technology](#)

From

To

☐ Include nearby airports ☐ Include nearby airports

Leave
 

Return
 

Adults

Class

Stops
☐ Non-stop only

☐ Search hotels too (in a new window)

TRAVEL TOPICS



Canada parks celebrate 100 years

TOP SLIDE SHOWS

- [How to break the law in 10 countries](#)
- [14 bizarre museums](#)
- [Kingly palaces around the world](#)
- [Try these affordable destinations](#)
- [World's oddest theme parks](#)

[All slide shows](#) | [Articles](#) | [Blog](#)

FLEXIBLE? COMPARE DATES AND DESTINATIONS

Find the dates with the lowest fares



Explore multiple routes at once



SPONSORED LINKS

LAST MINUTE FLIGHT DEALS

-  [\\$254 Chicago to Las Vegas](#)
-  [\\$260 Miami to Las Vegas](#)
-  [\\$255 Seattle to Las Vegas](#)

票价趋势预测系统 (Farecast)

- 由华盛顿大学Oren Etzioni教授在2003年创建
- 2008年被Microsoft Bing收购
- 截至2012年, Farecast利用了十万亿条价格记录
- 准确度已经高达75%, 平均每张机票可节省50 美元

票价趋势预测系统 (Farecast)

7-Day Low Fare Prediction



Tip: Wait

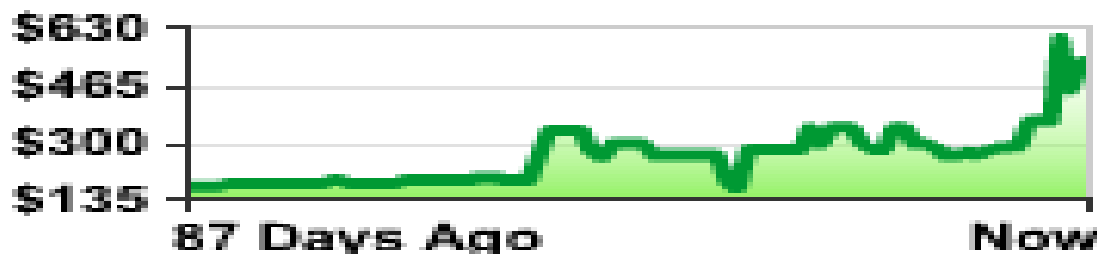
Fares Dropping or Steady.

Confidence: 70%

[Details](#)

Catch Fare Drop

Daily Low Fare History

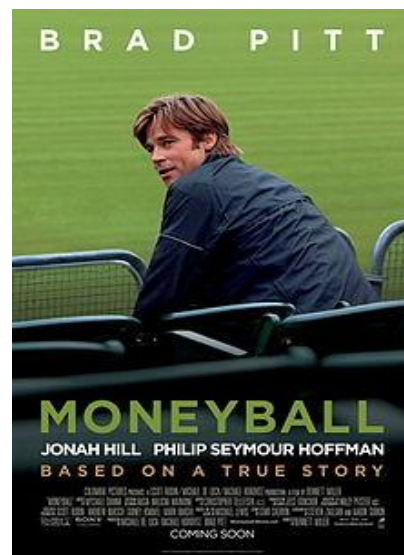


球员探寻系统

➤ Moneyball: The Art of Winning an Unfair Game

魔球——逆境中致胜的智慧

- 奥克兰市运动家棒球队的总经理Billy Beane采用统计分析模型对所有球员及相关信息进行分析
- 预测球员潜能及购买球员



美国总统大选 (2012)



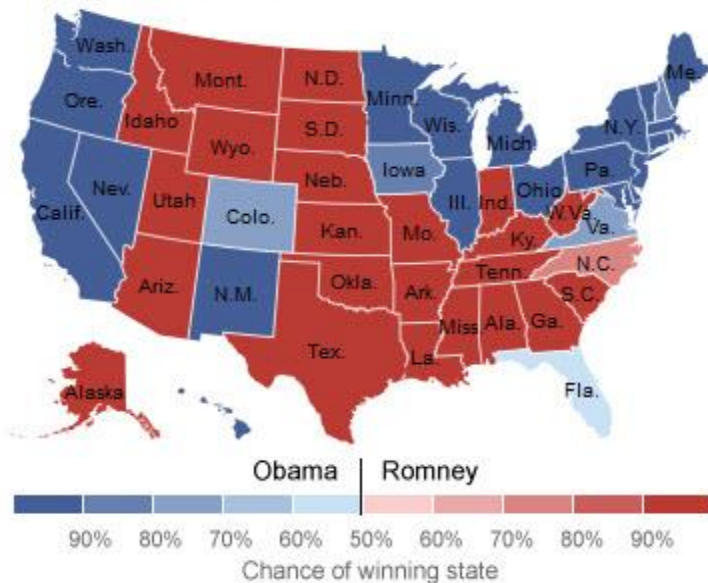
Romney 罗姆尼



Obama 奥巴马

当时媒体报道两人势均力敌

State-by-State Probabilities



奥巴马竞选团队分析Facebook, Twitter等文本数据
每天晚上运行 66000次模拟竞选
采取针对性措施, 如

- (1)帮助候选人选择最佳的演讲地点,日程及内容
- (2)以哪些区域为目标以及应当忽视哪些人
- (3)为吸引中立派女性选民,聘请乔治·克鲁尼来做广告



Nate Silver 通过数据分析预测奥巴马有86%
机会获胜, 而且准确预测出所有50个州的选举结果

课堂问题

1. 大数据时代下你认为有用(有趣)的应用有哪些并解释原因。注意这些应用最好是课堂上没有涉及到的。

2. 对于一般人来说，如何能获取对应问题1的真实数据？

注明：

(1) 答案； (2)姓名； (3) 学号； (4) 专业
